Router → MQTT EC2 through the VPN

Keep the broker at **52.64.103.203:3003/TCP**. Carry the router's connection inside OpenVPN to **3.24.63.90:1197/UDP**, then forward it to the broker.

Proposed configuration · deployment not performed by this document

1 STM router

Management: 192.168.10.1:8081 • VPN: 10.87.0.2 (previously observed)

MQTT destination stays 52.64.103.203:3003/TCP

Host route: 52.64.103.203/32 → tun-router



Encrypted OpenVPN over LTE / Internet

Outer transport: UDP 1197 • Inner connection: MQTT TCP 3003

2 VPN EC2

Public: 3.24.63.90:1197/UDP • Tunnel: 10.87.0.1 / tun-stm32

Decrypt → allow TCP 3003 forwarding → source NAT



TCP 3003 to 52.64.103.203

This leg needs MQTT TLS or a separate protected network.

3 MQTT EC2 — 52.64.103.203:3003/TCP

Broker sees the VPN EC2 public IP as the client's network source.

> **Two ports, two jobs.** UDP **1197** transports the encrypted VPN. TCP **3003** is the MQTT broker connection carried inside it. The broker does not move to port 1197.

Replies travel back through the VPN EC2, which reverses the NAT translation and sends them through the tunnel to the router. The router's MQTT host, port, credentials and topics can remain as configured.

IMPLEMENTATION RESPONSIBILITIES & VERIFICATION

What changes, and where?

Location	Required configuration
STM router 192.168.10.1	Enable the existing OpenVPN profile for **3.24.63.90:1197/UDP**. Confirm a tunnel address in **10.87.0.0/24**. Persist a host route for **52.64.103.203/32** through the VPN. MQTT continues to use **52.64.103.203:3003**.
VPN EC2 3.24.63.90	Keep the UDP **1197** listener reachable from the router's current public address. Enable IPv4 forwarding. Allow the router's tunnel address to reach **52.64.103.203 TCP 3003**, permit established replies, and apply source NAT on the outbound connection. Persist the rules and review EC2 source/destination checks for this NAT role.
MQTT EC2 52.64.103.203	Keep the broker listening on TCP **3003**. Allow TCP 3003 from the VPN EC2's public IP in the security group and host firewall. After testing and accounting for other clients, restrict broader public access to 3003. Preserve the separate TCP **3002** listener and access rules if other devices need them.

Make the router route persistent

The OpenVPN client configuration needs:

```
route 52.64.103.203 255.255.255.255 vpn_gateway
```

The router generates `/etc/router-web/openvpn/client.conf`; add the route to its persistent settings/configuration generator so later saves do not remove it. The current client uses `route-nopull`, so a route pushed only by the server is insufficient.

A /32 route covers every port on this one destination IP. Restricting routing to TCP 3003 alone requires policy routing. Ordinary Internet destinations continue to use the existing default route.

Verify the complete path

1. **Connect VPN:** check router and server logs, tunnel addresses and successful tunnel traffic. Check UDP 1197 access if authentication never completes.
2. **Check routing:** on the router, `ip route get 52.64.103.203` should select `tun-router`.
3. **Reconnect MQTT:** restart/reconnect the MQTT client so the test uses a new TCP connection. Observe TCP 3003 on the VPN server's tunnel and outbound interfaces.
4. **Confirm delivery:** verify the broker sees the VPN EC2 public IP and receives a known test message. Where applicable, verify a subscribed reply reaches the router.
5. **Check persistence:** reconnect the VPN and reboot the router; confirm MQTT resumes using the tunnel. Test the intended behaviour when the VPN is down.

> **Security boundary:** OpenVPN protects the router-to-VPN-server leg. It does not automatically encrypt the onward connection to the broker. A port number alone does not prove TLS is enabled. For VPN-only MQTT, block direct router-to-broker TCP 3003 outside the tunnel; otherwise the normal route may be used after VPN failure.

Source notes: project `docs/openvpn.md` and `web/openvpn_client.py`; router/VPN/MQTT discussion of 9 November 2026. The VPN profile was reported disabled in the latest discussion; older project records show an earlier successful connection.

References: OpenVPN manual — routes and route-nopull; AWS — NAT instances and source/destination checks.